

two voiced sounds are hardly distinguishable if only time-domain representation.

- 1) Why frequency-domain approach
- phân tích TH trong miền tần số
 - other characteristic of signals (which are not exhibited on time domain representation)
 - other characteristics of systems
 - especially LTI system:
 - response of a LTI system to a sinusoid is a sinusoid with same frequency.
 - different amplitude and phase:

2) Ideas / any signal can be described as a sum or integral of sinusoidal signals.

a) Continuous-time sinusoids:

$$x_a(t) = A \cos(2\pi f_0 t + \theta) \quad -\infty < t < \infty$$

A : amplitude
 θ : phase (radians) angular frequency (rad/s): $\Omega_0 = 2\pi f_0$
 f_0 : frequency (Hz)

b) Feature:

- $-\infty < t < \infty$
- 1) $F = \text{const} \Rightarrow T_p = \frac{1}{F} \Rightarrow x_a(t + T_p) = x_a(t)$
- 2) $F_1 \neq F_2 \Rightarrow$ Different signals
- 3) $f_0 \uparrow \rightarrow$ increase speed $\Leftrightarrow \uparrow$ num. of T
 t kien truong $\rightarrow$ may $\uparrow F$ den v.s cung.

c) Relation with complex exponentials.

$$A \cos(\Omega_0 t + \theta) = \frac{A}{2} e^{j\theta} e^{j\Omega_0 t} + \frac{A}{2} e^{-j\theta} e^{-j\Omega_0 t}$$

a) Harmonically related complex exponentials

$$s_k(t) = e^{j\Omega_k t} = e^{j2\pi f_k t} \quad k = 0, \pm 1, \pm 2, \dots$$

$s_k(t)$ is the k^{th} harmonic

b) Feature:

a) Discrete-time sinusoids

$$x[n] = x(nT) = A \cos(2\pi f_0 nT + \theta) = A \cos\left(2\pi \frac{f_0}{F_s} n + \theta\right)$$

or

$$x[n] = A \cos(2\pi f_0 n + \theta) = A \cos(\omega_0 n + \theta)$$

Normally frequency: $f \triangleq \frac{F}{F_s} = FT$

Normally angular frequency $\omega \triangleq 2\pi f = 2\pi \frac{F}{F_s} = \Omega T$

b) Properties:

$x[n] = A \cos(2\pi f_0 n + \theta)$ is periodic $\Leftrightarrow \exists ! k \mid f_0 = \frac{k}{N}$ two prime numbers

3) Extract form of the representation depends on

- signal is continuous-time | discrete-time
- periodic | aperiodic

	Periodic	Aperiodic
CT	CTFS Chuẩn Fourier	CTFT Chuẩn Fourier
DT	DTFS Chuẩn Fourier	DTFT

1) CTFS (Continuous Time Fourier Series)

a) Fourier Synthesis Equation

$$x(t) = \sum_{k=-\infty}^{\infty} c_k e^{j\Omega_k t}$$

Fourier Analysis Equation

$$c_k = \frac{1}{T_0} \int_{T_0} x(t) e^{-j\Omega_k t} dt$$

analyzes (break-up) a periodic signal $x(t)$ into a set of harmonic components

$$\{c_k\}: \text{Fourier series coefficients } (k=0, \pm 1, \pm 2, \dots) \quad P_{av} = \frac{1}{T_0} \int_{T_0} |x(t)|^2 dt = \sum_{k=-\infty}^{\infty} |c_k|^2$$

$$c_k = |c_k| e^{j\angle c_k}$$

$|c_k|$: magnitude spectrum of $x(t)$

$\angle c_k$: phase spectrum of $x(t)$ power spectrum:

Parseval's relation

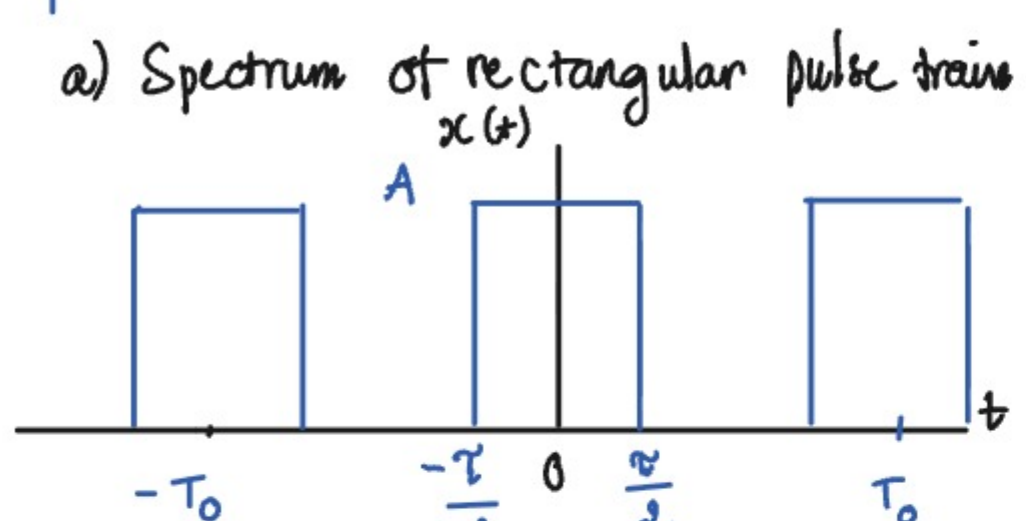
portion of the average power of signal $x(t)$

$\rightarrow$ contributed k^{th} harmonic components

the plot of $|c_k|^2$ as a function of frequency $F = kF_0$

$\rightarrow$ power is distributed at a set of discrete frequencies

$\rightarrow$ discrete or line spectrum with uniform spacing $F_0 = \frac{1}{T}$ (F_0 = fundamental frequency of $x(t)$)



$$c_k = \frac{1}{T_0} \int_{-\tau/2}^{\tau/2} A e^{-j\Omega_k t} dt = \frac{A}{T_0} \left[\frac{e^{-j\Omega_k t}}{-j\Omega_k} \right]_{-\tau/2}^{\tau/2}$$

$$= \frac{A}{T_0} \cdot \frac{e^{j\Omega_k \tau/2} - e^{-j\Omega_k \tau/2}}{-j\Omega_k}$$

$$= \frac{A\tau}{T_0} \cdot \frac{\sin(\Omega_k \tau/2)}{\Omega_k \tau/2} \quad k = \pm 1, \pm 2, 0, \dots$$

$$(c_0 = \frac{A\tau}{T_0})$$

a) Spectrum of rectangular pulse train: $T_0 = 5s \Rightarrow F_0 = \frac{1}{T_0} = 0.2 \text{ Hz}$
 $\tau = 1s \Rightarrow A = 1$

Ap dụng phân tích cấp CTFS, các hệ số phổ của $x(t)$ là:

$$c_k = \frac{1}{T_0} \int_{T_0} x(t) e^{-j\Omega_k t} dt = \frac{1}{T_0} \int_{-\tau/2}^{\tau/2} e^{-j\Omega_k t} dt$$

$$= \frac{1}{T_0} \left[\frac{e^{-j\Omega_k t}}{-j\Omega_k} \right]_{-\tau/2}^{\tau/2} = \frac{A}{T_0} \cdot \frac{1}{-j\Omega_k} \left(e^{-j\Omega_k \tau/2} - e^{j\Omega_k \tau/2} \right)$$

(Thay $T_0 = 5, F_0 = 0.2 \Rightarrow T_0 F_0 = 1$)

$$= -\frac{A}{\Omega_k 2\pi} \left(e^{-j\Omega_k \tau/2} - e^{j\Omega_k \tau/2} \right)$$

$$= -\frac{A}{\Omega_k 2\pi} \cdot -2j \sin(\Omega_k \tau/2)$$

$$= \frac{A}{\Omega_k} \cdot \sin(\Omega_k \tau/2) = A F_0 \tau \cdot \frac{\sin(\pi k F_0 \tau)}{\pi k F_0 \tau}$$

$$= A F_0 \tau \cdot \text{sinc}(k F_0 \tau) = 1 \cdot \frac{1}{5} \cdot \text{sinc}\left(k \frac{1}{5} \cdot 1\right) = \frac{1}{5} \text{sinc}\left(\frac{k}{5}\right)$$

Phổ biên độ của $x(t)$: $|c_k| = \frac{1}{5} \left| \text{sinc}\left(\frac{k}{5}\right) \right|$ do $c_k \in \mathbb{R} \forall k$

$$e^{jx} - e^{-jx} = 2j \sin x$$

$$f(x) = \text{sinc}(x) = \frac{\sin(\pi x)}{\pi x} = \begin{cases} 1 & (x=0) \\ 0 & (x=\pm 1, \pm 2) \\ |f(x)| \rightarrow 0 & \text{ khi } |x| \rightarrow +\infty \end{cases}$$